

SAURABH RAJ SHEKHAR

+91 96435 49110 | shekharsaurabhraj@gmail.com | [Linkedin](#) | [Github](#) | [Portfolio](#)

EDUCATION

Course	College / University	Year	CGPA / %
B.Tech (CSE – Data Science)	Netaji Subhas University of Technology	2024 – Present	7.36

SKILLS

Languages	Python, JavaScript, HTML, CSS, C++
Frameworks	React.js, Node.js, Express.js, Tailwind CSS, FastAPI, Flask
ML / Data	XGBoost, SHAP, GeoPandas, NumPy, Pandas, Scikit-learn
Tools	Git, VS Code, Postman, Jupyter Notebook, Vercel, Netlify, Render
Soft Skills	Quick Learner, Communication, Team Collaboration, Adaptability

PROJECTS

- **UrbanHeat – Urban Heat Island Analyzer**
Tech Stack: Python, FastAPI, XGBoost, SHAP, GeoPandas, React, Vite, Deck.gl, MapLibre GL, Recharts
 - AI/ML full-stack web app analyzing urban heat islands across 20 Indian cities using Landsat satellite data; single XGBoost model trained with leave-cities-out cross-validation for generalization to unseen cities.
 - SHAP-based driver analysis explaining per-grid heat contributors; interactive Scenario Builder simulates cooling interventions (tree cover, reflective roofs) with live before/after split-view map.
 - Submitted to ISRO BAH 2026 hackathon. Deployed on Vercel (frontend) and Render (FastAPI backend); GeoParquet flat files for zero-infrastructure data storage.
- **GitHub Radar – GitHub Contribution Analyzer**
Tech Stack: React, Vite, GitHub REST API, Recharts, Tailwind CSS
 - Dashboard that fetches and visualizes any user's GitHub profile data — repositories, commit history, language distribution, and contribution streaks — via the GitHub REST API.
 - Interactive Recharts visualizations for language breakdown and activity graphs; clean Apple-inspired UI built with React and Tailwind CSS.
- **Cryptex – Token-Based File Sharing Platform**
Tech Stack: Node.js, Express.js, MongoDB Atlas, Supabase Storage, HTML, CSS, JavaScript
 - Secure file sharing platform with upload, download, folder organization, and file preview — generating unique share tokens per item so database IDs are never exposed to end users.
 - Public/private visibility toggle; 9 REST API modules (files, folders, token access, visibility, download, preview); MongoDB for metadata, Supabase Storage for file content.
 - Deployed on Render with production-grade environment configuration and ISC-licensed open source release.
- **Automata Lab – Interactive Automata Visualizer**
Tech Stack: JavaScript, HTML, CSS (Vanilla — zero dependencies)
 - Web-based simulation tool for visualizing DFA, NFA, and model conversions — making Theory of Automata concepts interactive with step-by-step input processing and state transition diagrams.
 - Supports NFA-to-DFA conversion with live diagram rendering; deployed on Netlify with zero build tooling overhead.
 - Result: Widely used by NSUT CSDS peers for TAFL coursework practice.

OTHER INFORMATION

- Languages Known: Hindi, English
- LeetCode Profile: Zephyrex_21